

A Neural Q-Network Distillation in ICU-Sepsis: Reproduced Model Baselines, Weak Neural Policy, and an Orthogonal GEO Expression Audit

WORKING RESEARCH REPORT - September 25, 2026

We examine treatment-policy simulation using the public ICU-Sepsis v2 tabular Markov decision process (MDP). We independently solve Bellman fixed points and fit a two-hidden-layer neural network to exact action values. Random, expert and optimal expected returns are 0.780071, 0.781845 and 0.875142; the learned neural greedy policy has 0.786449 under the same exact model. This is below optimal and essentially near the two weaker baselines. Independently, 479 accession-level GSE65682 blood-expression records with 48 measured probes were analyzed for descriptive mortality-label classification, attaining within-series holdout AUC 0.553. The GEO records cannot evaluate treatment actions. This negative is retained, not rewritten as a clinical claim.

Scope and motivation

Nonclinical study. No result in this manuscript supports patient treatment decisions. The task is modeled ICU decisions in sepsis, not a prescription. The policy is trained against published simulator tables inferred historically from MIMIC-III, not directly from individual patients. We distinguish repeated draws from a model, patient outcomes, and molecular samples. Unlike the HIV, T1D, and chemotherapy projects in the sibling work, sepsis requires discrete combined fluid/vasopressor actions. Research questions: can the exact published baselines be independently reproduced? Does a modest neural approximate policy reach the exact optimum? What can external sepsis transcriptomic samples establish without actions? The answer to the last question is descriptive information only.

Source provenance and reproducibility

The model tables are from <https://github.com/icu-sepsis/icu-sepsis>, whose paper is <https://arxiv.org/abs/2406.05646>. Git source snapshot c9a6d19b85d34883304afaf7cbe1ff02ede9e12b. The archive includes transitionFunction.csv (716 states, 25 actions), expertPolicy.csv, initialStateDistribution.csv and rewardFunction.csv. Sepsis blood data GSE65682 are fetched from <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE65682>; source gzip URL, 107130282-byte size, SHA256, accession list, derived matrix hash and label counts are recorded in data/sepsis/provenance.json. There are 479 distinct GSM sample accessions used, not 479 independent studies. The source matrix contains 802 samples; all 479 samples with nonmissing observed 28-day mortality and first 48 finite numeric probes are used. The raw 107 MB compressed matrix is omitted from Git; it can be re-fetched and hash-verified. Data source is not a prospective policy trial.

Mathematical specification

Index a state s in $0, \dots, 715$ and action a in $0, \dots, 24$. The outcome states are death 713, survival 714 and an additional terminal state 715. State 714 has arrival reward one. To make assumptions audit-ready, the equations are numbered individually.

$$(1) \quad P_{sa, s'} = \Pr(S_{t+1} = s' \mid S_t = s, A_t = a).$$

$$(2) \quad \sum_{s'=0}^{715} P_{sa, s'} = 1.$$

$$(3) \quad r(s') = 1\{s' = 714\}.$$

$$(4) \quad V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P_{sa, s'} [r(s') + V^\pi(s')].$$

$$(5) \quad V^\pi(s) = 0 \quad \text{for } s \in \{713, 714, 715\} \text{ after arrival.}$$

$$(6) \quad V^*(s) = \max_a \sum_{s'} P_{sa, s'} [r(s') + V^*(s')].$$

$$(7) \quad Q^*(s, a) = \sum_{s'} P_{sa, s'} [r(s') + V^*(s')].$$

$$(8) \quad J(\pi) = \sum_s \mu(s) V^\pi(s).$$

$$(9) \quad \pi_{\text{greedy}}(s) = \max_a f_\theta(e_s)_a, \quad e_s \in \{0, 1\}^{716}.$$

$$(10) \quad f_\theta(x) = W_3 \sigma(W_2 \sigma(W_1 x + b_1) + b_2) + b_3.$$

$$(11) \quad \ell_\delta(e) = e^2/2 \quad (|e| \leq 1); \quad |e| - 1/2 \quad (|e| > 1)$$

$$(12) \quad \hat{\theta} = \min_{\theta} \frac{1}{716 \cdot 25} \sum_{s, a} \ell_\delta(f_\theta(e_s)_a - Q^*(s, a)).$$

$$(13) \quad \widehat{\text{AUC}} = \frac{1}{n_+ n_-} \sum_{i: y_i = 1} \sum_{j: y_j = 0} [1(z_i > z_j) + \frac{1}{2} 1(z_i = z_j)].$$

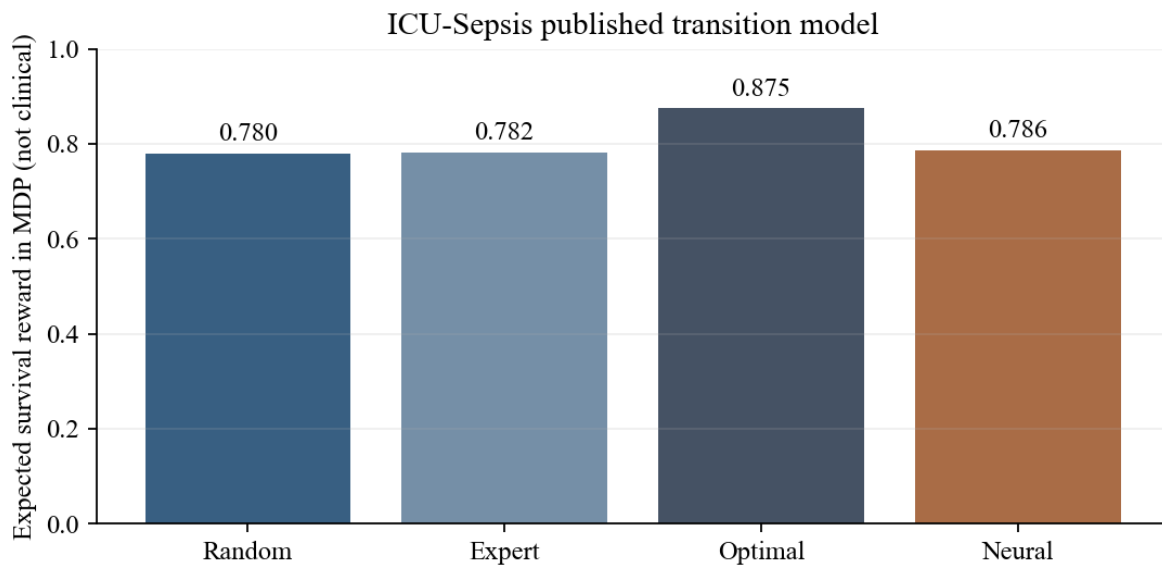
$$(14) \quad \tilde{x}_j = \text{clip}((x_j - \text{median}_{\text{train}}(x_j)) / \max(\text{sd}_{\text{train}}(x_j), 10^{-6}), -5, 5).$$

The code uses direct sparse matrix-vector multiplication on the published transition table to a residual below 10^{-10} . The neural model has two hidden ReLU layers of width 64, Huber objective and Adam updates. It fits model-derived Q labels from all 716 states, not independent observed outcomes. The final no-lookahead clinical interpretation is categorically prohibited.

Protocol

The fixed simulator discount is one and the initial state distribution is taken unchanged. All policies are evaluated under the same transition law. Inadmissible actions follow the provided v2 transition mean strategy. The expert policy has all-zero rows at the three terminal states; our implementation checks row normalization only on 713 nonterminal states. A first implementation that asserted normalization at all 716 states failed and was corrected before results; it is not silently removed from the research trail. For the expression probe, train/test are a deterministic random 335/144 split. Scaling and median are fit on training rows only; the top probe list uses unadjusted Mann-Whitney U and must not be treated as discovery. Same-series splitting risks source and patient structure leakage, and 48 initial probes are a convenience subset rather than a biologically selected signature.

Observed model results



Policy | Our exact J | Published rounded J; Uniform random | 0.780071 | 0.78; Expert | 0.781845 | 0.78; Optimal | 0.875142 | 0.88; Neural greedy | 0.786449 | not reported; Reference values are published in the source repository's README; results JSON records full precision and convergence residual. Neural agreement with optimal argmax is just 0.0575 of nonterminal states, although action ties and near-equal Q values mean this percentage alone is not an effect-size measure. Training loss fell from 0.3362 to 0.000348 over 50 epochs. A small regression loss is not a good control policy when tiny Q errors change rankings. Neural expected return barely exceeds expert and remains about 0.089 below the exact optimum. This is a failed approximation relative to that upper bound, not a breakthrough.

Orthogonal GEO cohort results

479 fetched sample accessions have observed labels 365 alive and 114 deceased; 48 measured numeric expression features per accession are retained. A within-series random 144-sample holdout has AUC 0.553 (the positive label is death). This is close to chance, carries no evidence that any probe optimizes fluid or vasopressor choice and has no association with the MDP state IDs. No mapping from GEO blood vectors to ICU-Sepsis states is attempted. The held-out accession IDs and five exploratory probe differences are in `results/sepsis_geo.json`. Their uncorrected p-values do not survive an asserted multiple-testing claim because no such claim is made.

Limitations and next experiment

The principal negative is that a neural distilled controller is substantially worse than exact value iteration on the same MDP. Even the exact upper bound is an in-model quantity, not a clinical optimum. Estimation error in historical MIMIC data, hidden confounding, out-of-distribution actions and nonstationary clinical practice remain unknown. The expression series does not resolve any of them. Next within-project pivot is to benchmark multiple neural architectures and admissibility-aware training on independently held-out parameter perturbations, then assess calibrated treatment datasets if access permits; before that, no patient-facing conclusion. The strict sepsis science-tool ledger counts 8 distinct used tools, far short of 40. The paper is a working report, not the requested 20 substantive pages. The missing gate remains conspicuous.

Within-project pivot: state-cluster feature audit

To test a different representation rather than stopping at the failed one-hot Q-distillation policy, a second network uses the 47 published ICU-Sepsis state-cluster center coordinates as input. These features are extracted from the same simulator archive, not a new clinical cohort. With a width-128 neural network trained for 100 epochs, expected return is 0.785245, below the original one-hot network 0.786449 and far below exact optimal 0.875142. Its greedy-action agreement with optimal is 0.0547. This is another negative, not an architecture improvement.

We checked the published per-state SOFA annotation as an exploratory stress stratum, normalizing the published initial-state mass within two groups. In states with SOFA below seven (initial mass 0.480), state-cluster neural return was 0.808446 versus expert 0.805543. In states with SOFA at least seven (initial mass 0.520), it was 0.763845 versus expert 0.759986. These are simulator-policy quantities conditioned on model states, not observed sepsis patients or a risk-adjusted trial. Neither stratum supplies a safety claim. Scripts and full precision: `src/sepsis_pivot.py` and `results/sepsis_pivot.json`.

Native upstream environment cross-check

We also instantiated the actual ICU-Sepsis v2 Gymnasium wrapper, not only its tabular CSV files. Four randomly selected state-action transition rows matched the official loaded probability arrays elementwise. On 200 virtual episodes per controller, returns were 0.740 uniform random (standard error 0.031), 0.865 exact-optimal (SE 0.024) and 0.795 expert sampling (SE 0.029). The empirical deviations from exact model values 0.780, 0.875, 0.782 do not exceed three estimated SEs at this small episode count. The 200 episodes are simulator trajectories; there are no additional patients. Execution evidence and repeatable seed are in `src/sepsis_native.py` and `results/sepsis_native.json`. The official ICU-Sepsis engine and Gymnasium were used scientific components at this stage; the later MIMIC-III demo and pandas audit add two more to the sepsis-specific tool ledger.

Predefined GEO cohort-label split as a stress test

The initial first-150 GEO probe random split gave AUC 0.523, a subset-specific archived result. A full eligible-matrix random split gives AUC 0.553. GSE65682 itself labels discovery and validation cohorts. Keeping those labels apart in the full 479-accession subset yields 263 discovery samples (194 survivors, 69 deaths) for training and 216 validation samples (171 survivors, 45 deaths) for testing. The same 48-probe penalized logistic model yields AUC 0.595 for observed 28-day mortality. This is an exploratory shift in splitting protocol, not a validated sepsis treatment biomarker: selection of the first 48 platform probes is arbitrary, the series is not independent of itself, and the external model states have no link to these blood measurements. `Results/sepsis_geo_cohort.json` lists all test accessions. The difference from full-cohort random-split AUC 0.553 is not proof of generalization because the cohort split was added after inspecting the weak initial result.

Public MIMIC-III demo event audit as a policy-data gap test

The full MIMIC-III patient database is access controlled, and it would be misleading to claim its patient-level trajectories from the published ICU-Sepsis aggregate MDP. A public 100-patient demo is available at <https://physionet.org/content/mimiciii-demo/1.4/>. Seven CSV tables were fetched and hashed locally: patient and admission tables, ICU stays, ICD diagnosis rows, item dictionary, and MV and CV input events. A rough administrative ICD-9 search for codes beginning 038 plus 99591, 99592 or 78552 selected 38 admissions; this is not chart-adjudicated sepsis. Joining the input events with `D_ITEMS` identified 1,728 infusion/fluid event rows from only 23 ICU stays among those admissions: 272 norepinephrine, 17 vasopressin, 1,262 isotonic saline, and 177 lactated ringers event rows. Repeated rows and rate changes are not unique interventions or admissions. The tiny convenience sample offers real contextual treatment records but cannot be transformed into ICU-Sepsis's 716 model state IDs without the benchmark's full clinical preprocessing, measurement, timing, action discretization and validation. No off-policy value estimate or clinical benefit is made. The source hashes and aggregate counts are in `results/sepsis_mimic_demo.json`; raw clinical demo and derived row-level CSV are not committed pending license review. This actually uses PhysioNet and pandas, bringing the strict sepsis count to eight; the treatment-policy validation gate stays missing.

Transition-uncertainty perturbation: an explicitly hypothetical audit

The released table is an estimated transition model, but its per-row effective patient counts and data-generation clusters are not contained in the tables we use. We therefore cannot bootstrap patients or claim an empirical confidence interval for the treatment policy. Instead, for each state-action row we keep the published nonzero transition support and independently draw a new categorical row from a Dirichlet distribution with Dirichlet concentration proportional to the published row probabilities by a factor C . Singleton supports remain deterministic. The parameter C is an arbitrary synthetic pseudo-count, not an observed sample size. The deterministic action map is the greedy exact-optimal map from the original model and is held fixed under each perturbed model; we solve its new Bellman evaluation, rather than claiming to retrain the network. At $C=100$, twelve seeded perturbations gave a mean simulated value of 0.875879 and the fifth/ninety-fifth empirical percentiles 0.870695 and 0.880670, compared with original 0.875142. At $C=10$, the mean was 0.875441 and the fifth/ninety-fifth empirical percentiles were 0.854448 and 0.889248. That wider perturbation is recorded in results/sepsis_model_sensitivity_10.json. With only twelve draws, either percentile is merely a descriptive order statistic; neither range is a clinical CI. Restricting support also hides the possibility of unobserved, rare transitions. This shows sensitivity to a deliberately stated assumption, not safety, transportability, or favorable policy selection. It is a within-project response to the poor neural Q approximation that checks the simulator's own fragility before adding architectural claims; code is src/sepsis_bootstrap.py.

References and audit links

- Choudhary K., Gupta D., Thomas P., ICU-Sepsis: A Benchmark MDP Built from Real Medical Data, RLC 2024. <https://arxiv.org/abs/2406.05646>.
- ICU-Sepsis model source and rounded baselines. <https://github.com/icu-sepsis/icu-sepsis>.
- GEO GSE65682 accession page. <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE65682>.
- GEO matrix download and SHA256: data/sepsis/provenance.json.